

Are LLMs Robust Enough for Operations Research?

An Adversarial Attack and Defense Study via AttackOR

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Abstract

Operations research (OR) solves complex real-world decision problems, from supply chain optimization to resource allocation, where errors carry real costs. LLMs are increasingly deployed to solve such problems by translating natural-language descriptions into optimization code. But how reliable are LLMs when problem descriptions are noisy, ambiguous, or deliberately misleading? We propose AttackOR, a unified framework for adversarial evaluation and robustness training in OR. We introduce four attack strategies that generate domain-consistent, instance-specific distractors to simulate the noisy and ambiguous inputs encountered in real-world OR: attribute injection, phantom options, arithmetic equivalence, and unit conversion. Experiments show that even a strong OR-tuned model is highly vulnerable, dropping from 59% to 40% Pass@1 under attack. To improve robustness, we combine supervised fine-tuning with GRPO-based reinforcement learning on adversarially augmented data. Our approach not only defends against adversarial inputs, achieving 71% Pass@1 under attack, but also improves performance on standard OR benchmarks, from 59% to 74%. Results reveal that clean-benchmark performance alone is insufficient to guarantee the reliability of LLM-based OR solvers in practice.

1 Introduction

Large language models (LLMs) have driven a transformative paradigm shift in operations research (OR), dismantling the long-standing expertise barrier that has limited the widespread adoption of optimization methodologies by enabling end-to-end translation of natural-language problem descriptions into solvable optimization formulations (Huang et al., 2024a; Ramamonjison et al., 2022). However, current LLM-based OR systems operate under a strong implicit assumption: the input problem descriptions are accurate, complete, and unambiguous. This assumption is often violated in practice. Real-world inputs may contain irrelevant context, outdated information, or semantically plausible statements that resemble valid constraints but should not affect the optimization task (Ramamonjison et al., 2023; Huang et al., 2024b). For example, Figure 1 shows a warehouse allocation task containing the statement, “Warehouse 2 is typically preferred for handling fragile goods.” Although non-binding, the model treats it as a hard constraint, assigns glass to Warehouse 2, and produces a suboptimal solution. This example shows that strong performance on clean benchmarks (Huang et al., 2024a; Ramamonjison et al., 2023; Huang et al., 2024b) does not guarantee robustness in realistic settings, where formulation errors can directly propagate to downstream decisions.

This robustness gap persists for two reasons. First, OR-specialized LLMs such as ORLM (Huang et al., 2024a) are trained and evaluated on clean, well-formed problem statements, which emphasize in-distribution accuracy rather than robustness to misleading inputs. Second, although adversarial robustness has been widely studied (Goyal et al., 2023), existing methods are designed mainly for general-purpose tasks rather than optimization-oriented reasoning. Lexical attacks (Jin et al., 2019; Li et al., 2020) and stress-testing or

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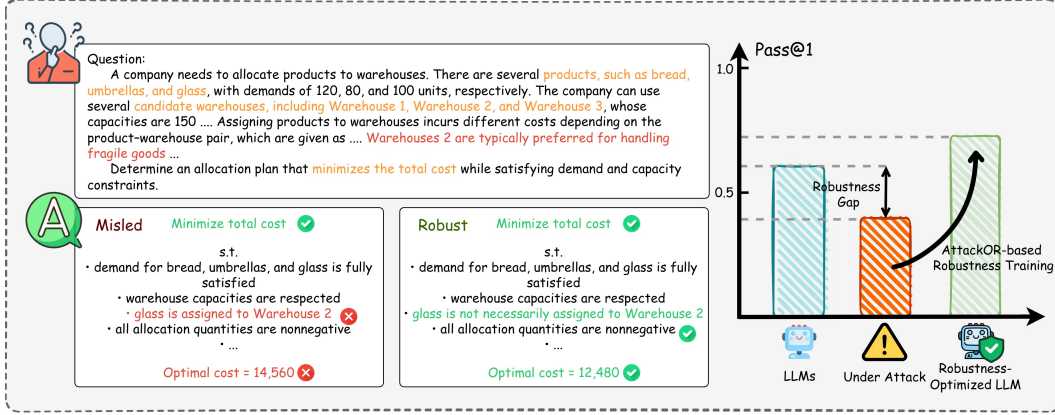


Figure 1: Robustness challenge in LLM-based OR System. Left: the statement is misinterpreted as a hard constraint, leading to a suboptimal solution. Right: the performance drop under real-world attack, which can be substantially reduced through AttackOR-based robustness training.

counterfactual methods (Ribeiro et al., 2020; Wu et al., 2021) can generate plausible distractions at the sentence or semantic level. Yet these perturbations are typically generic, weakly grounded in the specific problem instance. To capture real-world deployment risks, attacks should be tightly aligned with domain semantics and problem-specific constraints, so that misleading information resembles valid optimization structures rather than obvious noise. This leaves adversarial robustness in LLM-based OR largely underexplored.

To address this gap, we introduce AttackOR, a unified framework for adversarial evaluation and robustness training for OR. AttackOR consists of two attack categories: realistic textual perturbations, including arithmetic equivalence and unit conversion, which target numerical and measurement-related vulnerabilities; misleading information injection, including attribute injection and phantom options, which may cause misidentification of constraints or option interpretation. Together, these attacks capture both numerical and semantic sources of brittleness in LLM-based OR systems.

Our attacks reveal a substantial robustness gap in current LLM-based OR systems, as illustrated in Figure 1 (right): under Attack, a strong OR-tuned LLM drops from 59% to 40% Pass@1. To address this, we further introduce a robustness training pipeline built on the AttackOR benchmark. By augmenting training with adversarial examples, we expose the model to semantically and numerically misleading inputs and encourage it to better distinguish essential constraints from irrelevant context. Empirically, the resulting model achieves 71% Pass@1 under attack. These results show that adversarially augmented training not only improves robustness under realistic attacks but also enhances standard OR performance.

Our contributions are as follows:

- We propose AttackOR that produces realistic and domain-consistent adversarial examples through both textual perturbations and injection-based attacks.
- We construct the AttackOR benchmark, comprising 5,240 adversarial examples, providing a concrete testbed for robustness evaluation in LLM-based OR systems.
- We demonstrate that AttackOR-based training substantially improves robustness under realistic attack.

2 AttackOR

AttackOR is a unified framework for adversarial evaluation and robustness training in LLM-based OR problem solving. It constructs attacked OR instances that preserve the underlying

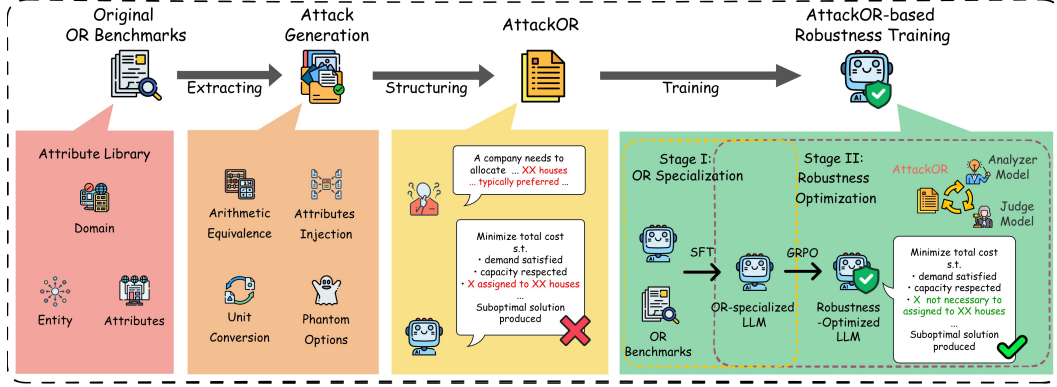


Figure 2: Overview of AttackOR, a unified framework for adversarial evaluation and robustness training in LLM-based operations research. AttackOR builds a data-derived attribute library and generates attack types that capture both semantic and numerical failure modes. Adversarial data is then used in a two-stage training pipeline.

optimization problem while introducing realistic distractions. The attack space consists of two kinds of attacks: realistic textual perturbations, including arithmetic equivalence and unit conversion, and misleading information injection, including attribute injection and phantom options. These generated attacked instances are used for both adversarial evaluation and robustness training. Figure 2 summarizes the overall pipeline.

2.1 Problem Formulation

We focus on solving operations research problems using large language models. Let x denote a natural-language OR instance, y the executable optimization code generated by the model, and $\pi_\theta(y | x)$ the code-generation policy parameterized by θ .

In the clean setting, the model is evaluated on original instances $x \sim \mathcal{D}$. Starting from a clean instance x , we further construct an attacked instance $\tilde{x} = \mathcal{T}(x; \delta)$, where $\mathcal{T}$ denotes a general transformation and $\delta \sim \Delta$ specifies an admissible perturbation. Here, admissible perturbations are semantically plausible text modifications that preserve the same underlying optimization problem.

The adversarial setting evaluates whether the model remains correct on attacked inputs $\tilde{x}$, namely, whether it can ignore non-operative distractors rather than translating them into operative constraints. Our method also uses the original-attacked pair $(x, \tilde{x})$ for both adversarial evaluation and robustness training. In experiments, we use Pass@1 on both clean and attacked inputs as the primary metric.

2.2 Data-Derived Attribute Library

To support semantically plausible adversarial generation, we introduce a data-derived attribute library that organizes OR-specific background knowledge into a lightweight domain-entity-attribute schema. We denote this library by

$$\mathcal{L} = \{(d, \mathcal{E}_d, \{\mathcal{A}_{d,e}\}_{e \in \mathcal{E}_d})\}_{d \in \mathcal{D}}, \quad (1)$$

where $\mathcal{D}$ denotes the set of domain categories. For each domain d , $\mathcal{E}_d$ denotes its set of salient entity types, and $\mathcal{A}_{d,e}$ denotes the set of plausible attributes associated with entity type e under domain d . Optional metadata is omitted for brevity.

Rather than manually enumerating distractors, we construct $\mathcal{L}$ from an OR corpus $\mathcal{C}$ through a semi-automatic pipeline. We identify recurring domains, collect the entity types that frequently arise within each domain, and then extract and normalize the attributes associated with each domain-entity pair, with limited manual curation where needed. The

resulting library is data-grounded and reusable across instances, rather than a hand-crafted list of distractor templates.

Importantly, the library does not introduce new operative constraints. Instead, it serves as a structured semantic prior for the library-guided attack families in AttackOR, enabling distractors that remain domain-consistent, entity-aligned, and realistic in context.

2.3 Attack Space and Construction

AttackOR defines adversarial construction as a family of admissible transformations that alter the surface form of an OR instance without changing its underlying optimization problem. Given an original instance x , an attacked instance is written as $\tilde{x} = \mathcal{T}_k(x; \delta_k)$, where $\mathcal{T}_k$ is the attack-family-specific instantiation of $\mathcal{T}$, $k \in \{\text{attr, phantom, arith, unit}\}$ denotes the attack family, and δ_k is the corresponding attack specification.

The four attack families fall into two categories. The first simulates realistic textual perturbations, including arithmetic equivalence and unit conversion, which target numerical and measurement-related vulnerabilities in OR reasoning. The second introduces misleading information into the problem description, causing the model to misidentify constraints, options, or other decision-relevant elements. This category includes attribute injection and phantom options.

Attack family instantiations

Attribute Injection. This attack inserts a plausible but non-operative attribute into the problem description, making irrelevant details appear actionable.

Phantom Options. This attack introduces an apparently attractive extra option together with conditions that apply only to that option, so that it appears competitive but is invalid under the original problem.

Arithmetic Equivalence. This attack rewrites quantities into mathematically equivalent forms, increasing the burden of quantitative interpretation without changing their values.

Unit Conversion. This attack expresses quantities in alternative but equivalent measurement units, preserving meaning while making unit-sensitive reasoning more difficult.

Together, these categories cover both realistic textual perturbations and misleading information injection. The same attack space is used both to construct adversarial evaluation instances and to generate robustness-oriented training data in AttackOR.

2.4 Robustness Optimizing Strategies

AttackOR enhances model robustness through a two-stage training procedure. Stage I establishes basic OR-solving capabilities, while Stage II performs robustness optimization on adversarially augmented data.

2.4.1 Stage I: OR-Specialized Supervised Fine-Tuning

We first obtain an OR-specialized base model by supervised fine-tuning on original OR data following the ORLM training paradigm (Huang et al., 2024a). This stage equips the model with the ability to translate natural-language OR descriptions into executable optimization programs and provides the initialization for subsequent robustness training.

2.4.2 Stage II: Interference-Aware Robustness Optimization

Starting from the OR-specialized base model, we further optimize the model on adversarially augmented data constructed from admissible perturbations. The objective is not only to recover the correct optimization outcome, but also to ignore non-operative distractors and resist misleading perturbations. A central challenge in this setting is that defense behavior

is difficult to reward directly, since semantically plausible distractions may mislead both the OR-specialized LLM and the reward judge.

Interference-Aware Defense Reward. To make defense behavior rewardable, we introduce an interference-aware reward pipeline. Given an original-attacked pair $(x, \tilde{x})$, a strong analyzer model produces a structured explanation

$$s = \mathcal{I}(x, \tilde{x}), \quad (2)$$

where s summarizes the distracting information, the mechanism by which it may mislead the OR-specialized LLM, and the intended defense behavior. This analysis is then provided as auxiliary context to a separate judge model when computing the defense reward:

$$R_{\text{def}} = J(y, \tilde{x}, s). \quad (3)$$

Here, J evaluates whether the generated output y correctly resists the attack, rather than encoding distracting information as operative constraints. This design makes otherwise ambiguous defense behavior directly rewardable.

Overall Reward and GRPO Optimization. We optimize the model with a weighted composite reward for adversarial OR solving. Let $R_{\text{format}}, R_{\text{exec}}, R_{\text{acc}}, R_{\text{def}} \in \{0, 1\}$ denote binary indicators for passing the code-format check, successful execution, objective-value correctness under a relative-error tolerance, and judge-based defense success in Equation 3, respectively. For clean rollouts, we set $R_{\text{def}} = 0$. Let $P_{\text{len}} \leq 0$ denote a progressive length penalty. The total reward is defined as

$$R_{\text{total}} = P_{\text{len}} + \alpha R_{\text{format}} + \beta R_{\text{exec}} + R_{\text{acc}} + R_{\text{def}}. \quad (4)$$

Here, α and β control the auxiliary rewards for valid formatting and successful execution, respectively.

We optimize this objective with GRPO (Shao et al., 2024), sampling multiple candidate responses for each attacked input and comparing them within a group. Since training instances are drawn from the same admissible perturbation space as evaluation, GRPO directly targets the failure modes exposed by AttackOR.

3 Experiments

3.1 Experimental Setup

Benchmarks and Attacks. We evaluate AttackOR on four representative OR benchmarks: IndustryOR (Huang et al., 2024a), NL4OPT (Ramamonjison et al., 2023), MAMO Easy/LP, and MAMO Complex/LP (Huang et al., 2024b), as shown in Table 1. Attacked instances are constructed using the four attack families of AttackOR, with a total of 5,240 adversarial samples.

Table 1: Benchmarks used in our experiments.

Benchmark	Description
IndustryOR (Huang et al., 2024a)	Realistic industry-oriented OR problems
NL4OPT (Ramamonjison et al., 2023)	Natural-language linear optimization
MAMO Easy/LP (Huang et al., 2024b)	Easier linear programming instances
MAMO Complex/LP (Huang et al., 2024b)	More complex linear programming instances

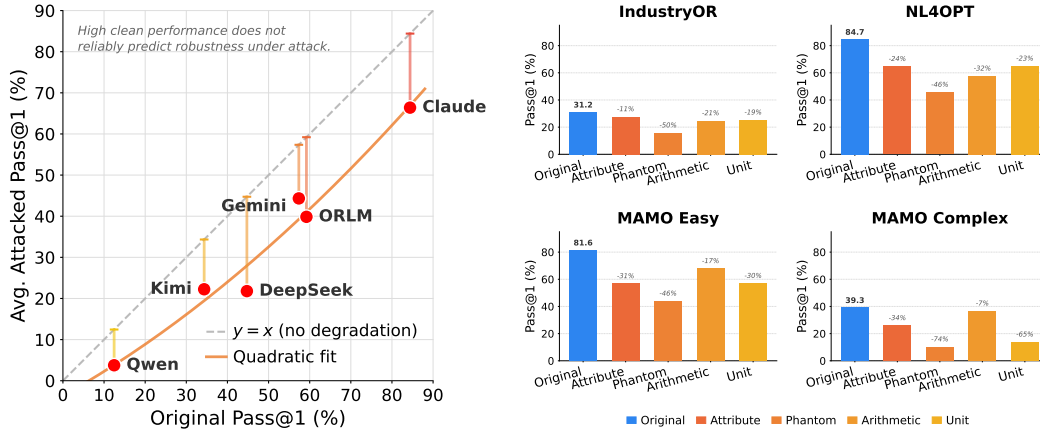
Models. We evaluate five frontier general-purpose models, Claude-Sonnet-4-6 (Anthropic, 2026), Gemini-3-Flash (Google DeepMind, 2025), DeepSeek-V3.2 (DeepSeek-AI et al., 2025), Kimi-k2.5 (Kimi Team, 2026), Qwen-Max (Yang et al., 2025a), and together with ORLM-LLaMA-3-8B (Huang et al., 2024a), an OR-tuned model designed for optimization-oriented reasoning and code generation. In addition, we use Qwen3.5-9B (Team, 2026) as the base model and construct two task-specific variants: Qwen3.5-9B-SFT and Qwen3.5-9B-AttackOR trained with the proposed adversarial robustness pipeline.

3.2 Vulnerability of LLM-based OR System Under Realistic Attacks

Table 2: Pass@1 (%) of representative LLMs under the original setting and attack families.

Model	Original	Attack Families				Avg. Attacked	Relative Drop
		Attribute	Phantom	Arithmetic	Unit		
Claude-Sonnet-4-6	84.40	56.86	68.75	67.10	72.87	66.40	-21.33%
Gemini-3-Flash	57.36	32.04	48.82	46.77	49.69	44.33	-22.71%
DeepSeek-V3.2	44.73	26.10	15.29	24.21	21.54	21.79	-51.29%
Kimi-k2.5	34.34	17.46	23.46	24.51	23.53	22.24	-35.24%
Qwen3-Max	12.45	4.91	1.94	7.48	0.72	3.76	-69.78%
ORLM-LLaMA-3-8B	59.21	43.69	28.86	46.66	40.19	39.85	-32.69%

We evaluate robustness by comparing model performance on clean and attacked instances across the selected OR benchmarks. The results show that strong clean performance does not reliably imply robustness to realistic attacks, as reported in Table 2 and Figure 3a. All evaluated models degrade substantially under perturbations, with relative drops ranging from 21.33% to 69.78%. This pattern holds for both frontier general-purpose models and the OR-specialized ORLM-LLaMA-3-8B, indicating that standard OR benchmarks are insufficient to evaluate robustness in realistic settings. Model vulnerability nevertheless varies markedly: Qwen-Max and DeepSeek-V3.2 are the most fragile overall, while Claude-Sonnet-4-6 and Gemini-3-Flash are comparatively more robust. The worst-case attack family also differs across models, suggesting that these attacks expose distinct vulnerabilities rather than a single shared weakness.



(a) Clean versus attacked performance across representative LLMs, showing clear degradation under attack.

(b) Performance of ORLM-LLaMA-3-8B under different attack families across datasets, indicating attack-specific and dataset-dependent failure modes.

Figure 3: AttackOR reveals robustness gaps in LLM-based OR systems.

Dataset-Dependent Attack Effects. We further evaluate the attack performance across the four benchmarks. Results show a clear dataset-dependent pattern, as shown in Figure 3b. Phantom consistently causes the largest drop, suggesting that plausible but invalid options are broadly disruptive. Beyond this shared trend, sensitivity varies substantially across datasets: IndustryOR is comparatively robust, MAMO Complex is especially vulnerable to Unit, and NL4OPT and MAMO Easy degrade strongly under all attack families. Overall, attack effectiveness depends on both the attack family and the dataset’s distribution and reasoning complexity.

Table 3: Effect of our robustness training across datasets and attack families. We compare SFT (Qwen3.5-9B-SFT) and AttackOR (Qwen3.5-9B-AttackOR), and report Pass@1 (%).

Dataset	Model	Original	Attack family				Avg. Attacked
			Attribute	Phantom	Arithmetic	Unit	
NL4OPT	SFT	90.91	60.00	53.12	58.82	80.00	62.99
	AttackOR	95.45	88.00	90.62	82.35	92.00	88.24
IndustryOR	SFT	33.33	23.08	25.00	38.46	16.67	25.80
	AttackOR	44.44	46.15	25.00	53.85	50.00	43.75
MAMO Easy	SFT	86.96	50.79	56.06	84.91	54.29	61.51
	AttackOR	89.86	87.30	83.33	88.68	92.86	88.04
MAMO Complex	SFT	50.00	44.44	16.67	41.18	13.04	28.83
	AttackOR	66.67	55.56	58.33	88.24	82.61	71.19
Avg.	SFT	65.30	44.58	37.74	55.84	41.00	44.79
	AttackOR	74.11	69.25	64.32	78.28	79.37	72.81

Generic irrelevant-sentence insertion causes a much smaller drop than AttackOR, suggesting that OR-specific admissible perturbations expose failures beyond general sensitivity to unrelated text. Details can be seen in Appendix A.3.3.

3.3 Robustness Improvement with AttackOR

Additionally, we assess whether AttackOR could improve robustness across attack families without sacrificing original OR performance. Starting from Qwen3.5-9B-SFT, we construct Qwen3.5-9B-CleanGRPO, Qwen3.5-9B-AttackSFT, and Qwen3.5-9B-AttackOR using clean-data GRPO, adversarial SFT, and the proposed robustness-training pipeline with adversarial data, respectively.

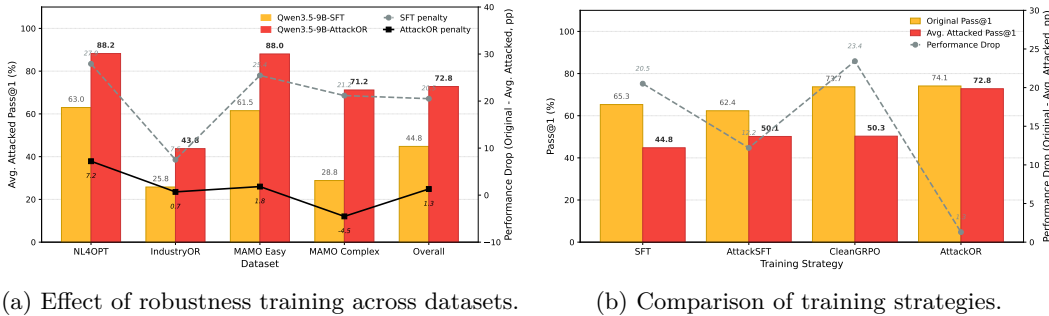


Figure 4: Robustness improvement of AttackOR. (a) Results across datasets. (b) Comparison with SFT, CleanGRPO, and AttackSFT.

AttackOR substantially improves robustness across datasets and attack families while also improving original OR performance, as summarized in Table 3. It is worth noting that prior to training, Qwen-3.5-9B had a Pass@1 of 0 for all OR problems, showing very limited OR performance. After two-stage training, Pass@1 increases from 0 to 65.30% and further to 74.11% at original benchmarks. As for average performance under attack, Pass@1 rises from 44.79% to 72.81%, with improvements observed across all four attack families. Moreover, the performance drop is markedly reduced across datasets, with the overall gap between original and attacked performance shrinking from 20.5 to 1.3 points, and even becomes negative on MAMO Complex, as shown in Figure 4a. These results indicate that AttackOR enhances robustness to both semantic and numerical perturbations without sacrificing, and in many cases improves original OR-solving ability. We also evaluate AttackOR on ORLM-LLaMA-3-8B, confirming the effectiveness of our pipeline on a different backbone; full results are provided in Appendix A.3.4.

To clarify the source of these robustness gains, we compare AttackOR with two control baselines that separately test the effects of additional GRPO optimization and adversarial data augmentation, as shown in Figure 4b. CleanGRPO improves Original Pass@1 from 65.30% to 73.71%, but raises Avg. Attacked Pass@1 only from 44.79% to 50.29%, leaving a performance drop of 23.42 percentage points. AttackSFT improves Avg. Attacked Pass@1 to 50.15%, but reduces Original Pass@1 to 62.35%. These results show that neither additional GRPO optimization nor adversarial data augmentation alone matches the robustness gains of AttackOR.

3.4 Generalization Across Datasets and Attack Families

To rigorously evaluate zero-shot robustness generalization, we partition our evaluation into seen (in-distribution) and unseen (out-of-distribution, OOD) subsets across two critical dimensions: OR datasets and attack families. We report the adversarial Pass@1 on seen-only, unseen-only, and their full union. This decoupling explicitly isolates intrinsic safety generalization from model improvements that merely overfit to training-covered perturbation templates.

Table 4: Generalization results on out-of-distribution datasets. AttackOR-Subset represents the model trained on a constrained distribution (specific scenarios or attack types).

(a) Dataset-level generalization.				(b) Attack-family generalization.			
Model	Seen	Unseen	All	Model	Seen	Unseen	All
SFT	39.21	61.51	44.78	SFT	41.11	55.84	48.89
AttackOR	67.73	88.04	72.81	AttackOR	70.05	75.34	73.06
AttackOR-Subset	56.21	78.58	61.80	AttackOR-Subset	70.71	70.70	71.46

The results in Table 4 indicate clear generalization across both OOD benchmarks and OOD attack families. Specifically, we set MAMO Easy as the held-out dataset in Table 4(a), and arithmetic equivalence as the held-out attack in Table 4(b). Note that the higher absolute Pass@1 scores on the OOD splits can be attributed to the inherently lower difficulty of the chosen held-out targets.

Crucially, even when trained with only partial coverage (AttackOR-Subset), the model substantially improves attacked Pass@1 over the SFT baseline on completely unseen splits (e.g., from 55.84% to 70.70% against unseen attacks). This demonstrates that our robustness improvements generalize beyond the exact training support, empowering the model to recognize underlying adversarial intents rather than memorizing surface-level patterns. While full-coverage training yields the strongest overall robustness, the subset results confirm non-trivial transferability in both dataset and attack dimensions.

Furthermore, we verify that the proposed safety alignment does not compromise general capabilities; additional visualizations, full experimental results, and the corresponding general-ability evaluation are provided in Appendix A.3.

3.5 Additional Evaluation and Ablation Analysis

Evaluation metrics. Beyond Pass@1, we report Avg. Pass@3, Avg. Pass@6, execution success, and the mean relative objective gap. Avg. Pass@ k averages performance over the original setting and the four attack families, while Pass@ k measures whether at least one of the first k sampled generations is correct. Execution success measures code executability, and the mean relative objective gap measures the relative deviation from the ground-truth objective value, with lower values indicating better numerical accuracy. Full results under these metrics are provided in Appendix A.3.7.

Stage-II component ablations. We further evaluate the two robustness-specific components of Stage-II training. The w/o R_{def} variant removes the defense-aware reward while retaining

the same adversarial training data and remaining reward terms. The w/o Analyzer variant removes the Analyzer-generated structured defense logic from the Judge.

Table 5: Ablation of the Stage-II robustness-training components. Higher values are better for Pass@ k and execution success, while a lower mean relative objective gap is better.

Model	Avg. Pass@1	Avg. Pass@3	Avg. Pass@6	Exec. Success	Mean Rel. Gap
Qwen3.5-9B-AttackOR	73.07	74.84	79.15	98.99	24.69
w/o R_{def}	64.05	73.45	76.60	98.40	62.88
w/o Analyzer	62.51	70.75	74.03	97.84	11.14

Removing R_{def} lowers Avg. Pass@1 from 73.07% to 64.05% and Avg. Pass@6 from 79.15% to 76.60%, while increasing the mean relative objective gap from 24.69 to 62.88, as shown in Table 5. These declines indicate that the defense-aware reward improves both robustness and numerical reliability. Removing the Analyzer further reduces Avg. Pass@1 to 62.51%, Avg. Pass@6 to 74.03%, and execution success to 97.84%. Although this variant yields a lower mean relative objective gap, its weaker Pass@ k and execution results show that Analyzer-generated defense logic provides a more informative reward signal than direct judging from the prompts and model output. Overall, both R_{def} and the Analyzer are important to the Stage-II robustness gains.

4 Related Work

4.1 Language Models for Operations Research

Recent work examines how LLMs can support operations research, from translating natural-language optimization problems into formal formulations and solver-ready code to broader modeling pipelines. Benchmarks and shared tasks such as NL4OPT (Ramamonjison et al., 2023), together with industry-oriented resources (Huang et al., 2024a) and modeling-oriented corpora and methods (Huang et al., 2024b), provide standardized evaluation settings for this problem. More recent efforts have expanded this line by revisiting dataset quality, introducing the LogiOR benchmark, developing expert-guided reasoning methods for optimization modeling (Yang et al., 2025b), and decomposing OR problem solving into modeling, code generation, and debugging stages through reasoning-centric agent frameworks (Zhang et al., 2025). Surveys further document rapid progress in data construction, training, inference, and evaluation protocols for optimization modeling with LLMs (Xiao et al., 2025). Despite this progress, evaluation is still conducted largely on original problem statements, where success is defined by the correctness of the constructed model and the generated solution under in-distribution inputs. By contrast, our setting focuses on reliability when the input remains plausible but includes misleading or irrelevant information, a challenge that standard OR benchmarks do not capture well.

4.2 Adversarial Attacks on Language Models

A large body of work studies adversarial attacks and defenses in NLP (Goyal et al., 2023), including gradient-based and search-based word-level attacks (Jin et al., 2019; Li et al., 2020) as well as attacks on aligned instruction-tuned models (Zou et al., 2023). Related security surveys further identify jailbreaks, poisoning, and privacy risks as major research directions for LLMs (Das et al., 2025). Most of these attacks are developed for classification or generic instruction-following settings, and typically rely on short prompts or surface-level perturbations. ReCode evaluates code-generation robustness using natural, semantics-preserving perturbations of problem descriptions and code-related elements (Wang et al., 2023). OR tasks impose stronger structural constraints: misleading content must remain locally plausible, interact with numerical values and units, and preserve the original decision variables, constraints, objective, and ground-truth solution. Accordingly, the attacks considered in our setting are better viewed as structured, instance-grounded distractions rather than imperceptible token-level edits (Goodfellow et al., 2014).

4.3 Robustness Training for Language Models

Robustness and alignment in LLMs are often studied in post-training settings. A standard formulation is reinforcement learning from human feedback, where PPO (Schulman et al., 2017) optimizes a policy against a learned reward model (Ouyang et al., 2022). GRPO reduces variance by comparing relative rewards within a group and has emerged as a practical alternative to PPO in large-scale LLM training (Shao et al., 2024). Recent work also explores the robustness of code-generation models under semantics-preserving input perturbations. ReCode provides a comprehensive benchmark for evaluating robustness across diverse perturbation types (Wang et al., 2023). CodeFort studies robust training for code-generation models through perturbation-based data augmentation and robustness-oriented objectives (Zhang et al., 2024). However, these approaches are usually instantiated with generic helpfulness or harmlessness rewards, rather than signals tied to executable optimization artifacts such as generated optimization code, feasibility, and objective value under semantically plausible distractors. Our setting therefore requires robustness rewards that are specific to OR reasoning and execution.

5 Limitations

Our study has several limitations that suggest directions for future work. Although the benchmark covers four semantically plausible and instance-grounded attack families, it still reflects a bounded attack space and does not capture all realistic failure modes in OR, especially those involving broader structural changes to the optimization problem. In addition, our formulation focuses on input-level robustness, leaving other failure sources in LLM-based OR systems, such as solver invocation, execution assumptions, and downstream interpretation, for future study. Finally, the current evaluation centers on single-turn responses under original and attacked inputs, and therefore does not address interactive settings where models may ask clarification questions, revise intermediate formulations, or recover through multi-turn feedback.

6 Conclusion

In this paper, we introduce AttackOR, a framework for evaluating and optimizing robustness in LLM-based operations research. Our results show that even strong OR-tuned LLMs, while performing well on clean benchmarks, remain highly vulnerable to perturbations in problem descriptions. To make this setting measurable, we constructed an AttackOR benchmark spanning four OR datasets and four complementary attack families. Experiment results reveal that strong performance on current benchmarks does not translate into reliable behavior under more realistic conditions.

We further showed that the robustness gap can be substantially reduced through AttackOR-based robustness training. Starting from an OR-specialized Qwen3.5-9B model, adversarial training improves performance broadly across datasets and attack types. These gains also generalize beyond the exact training coverage, suggesting that the model learns more robust optimization-relevant reasoning rather than merely memorizing specific perturbations. We hope AttackOR provides a useful insight for future work on robust and reliable LLM-based realistic applications.

Reproducibility Statement

To ensure the reproducibility of experimental results, we have provided comprehensive details of our experimental setup, methodology, and evaluation metrics throughout the paper and in the appendices. We also provide the source code in the supplementary material. Below we summarize several important aspects to facilitate reproducible results:

- **Experiments.** Detailed descriptions of the experimental environments are provided in Manuscripts. The full results are provided in main manuscripts and in Appendix.
- **Open Source.** We open our source code in the anonymous repository <https://github.com/HanxiZhu/AttackOR>.
- **Environment.** Experiments are conducted using high-performance computational resources: 8×NVIDIA H200–80GiB GPUs.

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A Appendix

A.1 Implementation Details

A.1.1 Attribute library.

To support realistic and domain-consistent perturbations, we maintain an attribute library organized as domain category $\rightarrow$ entity type $\rightarrow$ plausible attributes. Each attribute contains a short name, a description, a unit, and an example usage template. The library is constructed with an LLM-assisted bootstrapping procedure that first identifies domain categories and entity types from benchmark questions, and then generates both entity-specific and generic attributes.

-
- 1: Start from benchmark questions
 - 2: Identify domain categories
 - 3: for each domain do
 - 4: Identify common entity types
 - 5: for each entity type do
 - 6: Generate plausible entity-specific attributes
 - 7: end for
 - 8: Generate generic domain-level attributes
 - 9: end for
 - 10: Assemble the hierarchical attribute library
-

A.1.2 Prompt Templates

Library construction prompts.

Prompt for Domain Category Extraction

Purpose. Identify the most common domain categories from benchmark questions.

Prompt.

You are analyzing a collection of optimization problems to identify common domain categories.

Here are some sample questions from the dataset:

{QUESTIONS_TEXT}

Your task is to identify the most common domain categories (e.g., production, transportation, investment, scheduling, resource allocation) that appear in these optimization problems.

Output a JSON list of category names and their descriptions:

```
{
  "categories": [
    {
      "name": "production",
      "description": "problems involving factories, workers, manufacturing, production planning"
    },
    {
      "name": "transportation",
      "description": "problems involving trucks, warehouses, ports, logistics, shipping"
    }
  ]
}
```

Only include categories that are clearly represented in the sample questions.
Output ONLY the JSON, no explanations.

Prompt for Entity Type Extraction

Purpose. Identify the most common entity types within a specific domain category.

Prompt.

You are analyzing optimization problems in the "{CATEGORY_NAME}" domain: {CATEGORY_DESC}.

Here are sample questions from this domain:

{QUESTIONS_TEXT}

Your task is to identify the most common entity types (e.g., factory, worker, product, truck, warehouse) that appear in these problems.

Output a JSON list of entity names and their descriptions:

```
{
  "entities": [
    {
      "name": "factory",
      "description": "..."
    },
    {
      "name": "worker",
      "description": "..."
    }
  ]
}
```

Only include entities that are clearly represented in the sample questions.
Output ONLY the JSON, no explanations.

Prompt for Distractor Attribute Generation

Purpose. Generate realistic per-unit attributes to serve as distractors for an entity type.

Prompt.

You are an expert in the "{CATEGORY_NAME}" domain.

You need to identify realistic attributes that could be associated with "{ENTITY_NAME}" described as {ENTITY_DESC} in optimization problems.

These attributes should be:

1. Realistic industry standards or requirements
2. Per-unit attributes, not global constraints
3. Suitable as distracting information in optimization problems
4. Free of value ranges; only provide name, description, unit, and example usage

Output a JSON list of attributes:

```
{
  "attributes": [
    {
      "name": "quality certification level",
      "description": "...",
      "unit": "level",
      "example_usage": "Each {ENTITY_NAME} must maintain a quality certification level"
    },
    {
      "name": "safety training certification",
      "description": "...",
      "unit": "certification",
      "example_usage": "Each {ENTITY_NAME} requires safety training certification"
    }
  ]
}
```

Generate 3-5 realistic attributes.
Output ONLY the JSON, no explanations.

Prompt for Generic Attribute Generation

Purpose. Generate domain-level generic per-unit attributes for fallback use.

Prompt.

You are an expert in the "{CATEGORY_NAME}" domain.

Generate generic attributes that could apply to any entity in this domain.

These should be:

1. Generic enough to apply to various entities
2. Per-unit attributes, not global constraints
3. Realistic industry standards

Output a JSON list:

```
{
  "attributes": [
```

```

    {
      "name": "operating standard",
      "description": "...",
      "unit": "standard",
      "example_usage": "Each entity must meet operating standard requirements"
    }
  ]
}

```

Generate 2-3 generic attributes.
Output ONLY the JSON, no explanations.

Attack generation prompts.

Prompt for Domain and Entity Identification

Purpose. Identify the domain category and salient entities for a single OR instance.

Prompt.

Analyze the following optimization problem and identify:

1. The domain category
(e.g., production, transportation, investment, scheduling, resource allocation)
2. The main entities mentioned in the problem
(e.g., factory, worker, truck, warehouse)

Question:

{QUESTION}

Output JSON:

```

{
  "domain": "production",
  "entities": ["factory", "worker", "product"]
}

```

Output ONLY the JSON, no explanations.

Prompt for Distracting Attribute Selection

Purpose. Select the least relevant yet plausible attribute for attribute injection.

Prompt.

You are selecting a distracting attribute for an adversarial attack on an optimization problem.

Question:

{QUESTION_PREFIX}

Available attributes:

{ATTRIBUTES_TEXT}

Select the attribute that is:

1. Least relevant to solving the optimization problem
2. Still plausible and realistic
3. Potentially confusing for an AI solving this problem

CRITICAL: The selected attribute must NOT affect the ground truth answer or change the feasible region of the optimization problem.

Output JSON:

```

{
  "selected_attribute": {
    "name": "quality certification level",
    "description": "...",
    "unit": "level",
    "example_usage": "Each entity must maintain a quality certification level"
  }
}

```

Output ONLY the JSON, no explanations.

Prompt for Attribute Value Generation

Purpose. Generate a plausible attribute value consistent with the instance's numeric scale.

Prompt.

You are generating a distracting value for an attribute in an optimization problem.

Question context showing value ranges:
{QUESTION_PREFIX}

Attribute to generate value for:
- Name: {ATTRIBUTE_NAME}
- Description: {ATTRIBUTE_DESC}
- Unit: {ATTRIBUTE_UNIT}

Value range in question: {MIN_VAL} to {MAX_VAL}
Value format: {FORMAT_TYPE}

Generate a value that:
1. Has a similar magnitude to the values in the question
2. Matches the format: integer or decimal
3. Is realistic for this attribute
4. Is confusing but not obviously wrong

CRITICAL: The generated value and description must NOT affect the ground truth answer or change the feasible region of the optimization problem.

Output JSON:
{
 "value": 3.2,
 "unit": "level",
 "full_description": "Each entity must maintain a quality certification level of 3.2"
}

Output ONLY the JSON, no explanations.

Prompt for Natural Attribute Fusion

Purpose. Integrate the selected attribute into the problem statement fluently and naturally.
Prompt.

You are an industry expert adding a realistic requirement to an optimization problem.
Your task is to naturally integrate the following attribute information into the problem statement.

Original question:
{QUESTION}

Attribute to add:
{FULL_DESCRIPTION}

Instructions:
1. Act as an industry expert providing additional context
2. Integrate this attribute naturally by either:
 - inserting a new sentence, or
 - modifying an existing sentence
3. Use authoritative language such as:
 "According to regulations", "Industry standards require", "Compliance mandates"
4. Make it sound like a legitimate requirement
5. Ensure the sentence is grammatically correct and fluent

CRITICAL: The integrated attribute must NOT affect the ground truth answer or change the feasible region of the optimization problem.
It should remain a distracting per-unit property.

Output the complete modified question with the attribute naturally integrated.
Do NOT add explanations or markdown formatting.

Prompt for Scarce Resource Attribute Selection

Purpose. Select an attribute that can plausibly act as a scarce resource in a phantom-option attack.

Prompt.

You are selecting one attribute to use as a scarce resource in a phantom option constraint.
The constraint will state that this phantom option requires X, but X is 0 or unavailable.

Question:
{QUESTION_PREFIX}

Available attributes:
{ATTRIBUTES_TEXT}

Select the attribute that:

1. Is plausible as something that could be in short supply or unavailable
2. Fits the domain of the question
3. Sounds natural and convincing when stated as 0 or limited

Output JSON:

```
{
  "selected_attribute": {
    "name": "attribute name",
    "description": "description",
    "unit": "unit",
    "example_usage": "example usage"
  }
}
```

Output ONLY the JSON, no explanations.

Prompt for Phantom Option Generation

Purpose. Generate a seemingly dominant phantom option that mirrors the original instance format.

Prompt.

You are an adversarial expert. Create a super-optimized phantom option for this problem.

Original Question:
{QUESTION}

Requirements:

1. Mathematical dominance:
Identify the best existing option and create a phantom option that is 20-30 percent more efficient.
It should appear to be the mathematically ideal choice.
2. Contextual mirroring:
Match the exact phrasing structure of existing entities.
3. Parameter integrity:
Provide all coefficients so the model can build a full variable for it.
All values must remain plausible within the current scale.
4. No spoilers:
Output only the description of the new option.
Do not mention that it is fake or restricted.

Example logic:

If Product A makes 10 and Product B makes 12, make Product C make 15
with the same or fewer resources.

Prompt for Phantom Constraint Generation

Purpose. Generate one sentence that makes the phantom option infeasible via scarcity.

Prompt.

Write exactly ONE sentence stating that a certain option requires "{X_NAME}",
or a natural phrasing of it, but the available amount, quota, or allocation is 0 or strictly limited,
so that option cannot be chosen.

Attribute:

- Name: {X_NAME}
- Description: {X_DESC}
- Example usage: {X_USAGE}

{PHANTOM_CTX_OPTIONAL}

Requirements:

- One sentence only
- No preamble
- No "For example"
- No markdown
- The sentence must clearly make the option infeasible because X is 0 or unavailable

Output ONLY that one sentence.

Prompt for Phantom Option Fusion

Purpose. Integrate the phantom option and its infeasibility constraint into the original problem statement.

Prompt.

You are integrating a phantom option and its constraint into an optimization problem. Output the complete modified question.

Original question:
{QUESTION}

New phantom option to add:
{PHANTOM_OPTION_TEXT}

Constraint sentence to add:
{CONSTRAINT_SENTENCE}

Instructions:

1. Insert the phantom option in the same place and format as existing options
2. Insert the constraint sentence naturally
3. Ensure the constraint clearly rules out the phantom option
4. Do NOT change the ground truth answer or the feasibility of the real options
5. Output ONLY the complete modified question
6. No explanations and no markdown

Defense and auditing prompts.

Prompt for Defense Logic Extraction

Purpose. Summarize the correct defense logic for an attacked instance by comparing it with its baseline counterpart.

Prompt.

Compare the following two optimization problems.
Identify the specific interference injected in the attack question
and summarize the correct defense logic in one sentence.

Baseline Question:
{BASELINE_QUESTION}

Attack Question:
{ATTACK_QUESTION}

Please analyze:

1. What interference was injected in the attack question?
2. What is the correct defense logic to handle this interference?

Return only the one-sentence defense logic summary, nothing else.

Prompt for Defense-Logic Compliance Auditing

Purpose. Audit whether the model reasoning explicitly follows the required defense logic.

Prompt.

You are an OR auditing expert.

Defense Logic Required:
{DEFENSE_LOGIC}

Model Reasoning:
{MODEL_REASONING}

Did the model reasoning explicitly identify and correctly handle
the required defense logic?

Answer purely with [YES] or [NO].

A.2 Training Settings

Dataset splits. We convert both clean and attacked instances into a unified training format while preserving metadata, including attack type and a one-sentence defense rationale for attacked samples. The full pool is shuffled with a fixed random seed (42) and split into

train/validation/test sets with an 80/10/10 ratio. Unless otherwise stated, robustness training uses the training split, validation performance is used for monitoring, and ablations may restrict the training pool to specific sub types. For the out-of-distribution generalization settings in Table 4a, MAMO Easy is held out as the test dataset in (a), and arithmetic equivalence is held out as the test attack family in (b).

Stage I: supervised fine-tuning (SFT). We perform full-parameter fine-tuning of Qwen3.5-9B using DeepSpeed ZeRO-3 in bf16. Training runs for 3 epochs with a global batch size of 128 on 8 GPUs. We use AdamW with a linear learning-rate schedule and a 3% warmup ratio. The main-paper configuration uses a learning rate of 5×10^{-5} , selected from a small grid over learning rate, batch size, and epochs.

Stage II: AttackOR robustness training (GRPO). We further optimize the Stage I model with GRPO in VERL, using vLLM-based rollouts. Training runs for 15 epochs with a global batch size of 128 on 6 GPUs, with maximum prompt and response lengths of 3000 and 4000 tokens. The actor learning rate is 1×10^{-6} with 10% warmup. For each prompt, we sample $n=6$ rollouts and apply low-variance KL regularization with coefficient 0.1 and an entropy coefficient of 0.005. We use a PPO mini-batch size of 64 with per-GPU micro-batching for memory efficiency. For rollout serving, we use tensor parallelism of 2, cap GPU memory utilization at 0.8, and disable chunked prefill. For the composite reward in Equation 4, we set $\alpha = 0.2$ and $\beta = 0.3$ in all experiments.

Hardware, frameworks, and precision. All experiments are conducted on NVIDIA H200 GPUs. Stage I uses 8 GPUs and Stage II uses 6 GPUs, with a separate vLLM judge instance when needed. We use bf16 for SFT, VERL for GRPO, and vLLM for batched rollouts. The defense judge is served through an OpenAI-compatible interface for reward-time auditing. We use coptpy as the solver interface for executing the generated optimization programs.

Training cost and inference overhead. We further report the computational cost of the Qwen3.5-9B training stages in Table 6. The reported GPU cost corresponds to each training stage rather than the cumulative cost of obtaining the final model.

Table 6: Training cost of the Qwen3.5-9B variants. GPU cost is measured in H200 GPU-hours for the corresponding training stage.

Training Stage	GPU Cost (H200 GPU-h)
Stage-I SFT	~24
AttackSFT	~8
Stage-II AttackOR	~175

Stage-II AttackOR training takes approximately 25 hours, using six H200 GPUs for the policy model and one additional H200 GPU for the Judge service. Its higher cost mainly comes from GRPO rollouts, repeated solution sampling, execution feedback, and reward-time auditing. The Analyzer generates defense rationales offline before training, while the Judge is used only to construct the defense reward. Neither component is required after training, so Qwen3.5-9B-AttackOR is deployed as a standalone model with no additional inference-time overhead.

A.3 Additional Experiments

A.3.1 General Ability After Robustness Training

We test whether robustness optimization on the Qwen line harms broad capabilities. We compare Qwen3.5-9B and Qwen3.5-9B-AttackOR on MMLU (Hendrycks et al., 2020), GSM8K (?), and IFEval (?).

Table 7: General ability on the Qwen line before and after AttackOR robustness training. Each cell reports Pass@1 (%).

Model	MMLU	GSM8K	IFEval
Qwen3.5-9B	81.66	92.74	28.87
Qwen3.5-9B-AttackOR	76.39	88.63	28.94

Table 7 checks for catastrophic forgetting on general reasoning and instruction-following benchmarks. The intended pattern is improved OR robustness (Table 3) together with stable general Pass@1 relative to Qwen3.5-9B.

A.3.2 Cross-Model Robustness Comparison

Figure 5 shows the full cross-model robustness comparison. All models degrade under realistic attacks, indicating that strong clean performance does not reliably translate into robustness.

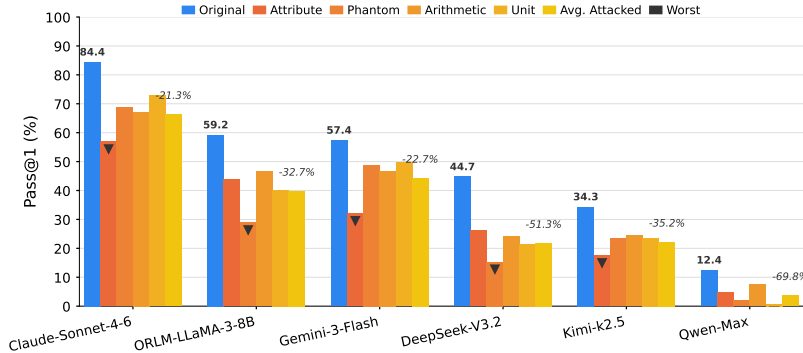


Figure 5: Cross-model robustness comparison under attacks. We report Pass@1 on the original inputs, each attack family, the average attacked setting, and the worst-case setting.

A.3.3 Comparison with Generic Irrelevant Context

To distinguish AttackOR from generic irrelevant-context perturbations, we construct a baseline that inserts fluent but unrelated sentences without OR-specific entity or attribute grounding. Unlike AttackOR, these sentences are not embedded in the same decision context and do not target specific OR modeling failures.

Table 8: Comparison between generic irrelevant-sentence insertion and AttackOR perturbations on Qwen3.5-9B-SFT. Performance drop is measured relative to the original problems.

Input Setting	Pass@1	Performance Drop (pp)
Original Problems	65.30	—
Generic Irrelevant Sentences	62.56	2.74
AttackOR Perturbations	44.79	20.51

Generic irrelevant-sentence insertion reduces Pass@1 by only 2.74 percentage points, whereas AttackOR causes a 20.51-point drop, as shown in Table 8. This large difference indicates that the observed vulnerability cannot be explained by unrelated text alone. Instead, domain-grounded perturbations that remain plausible within the OR scenario are substantially more difficult for the model to distinguish from decision-relevant information. These results support the need for OR-specific admissible perturbations when evaluating the robustness of LLM-based OR solvers.

A.3.4 Cross-Backbone Validation

To examine whether the effectiveness of AttackOR is specific to Qwen3.5-9B, we further apply the Stage-II robustness-training pipeline to ORLM-LLaMA-3-8B, an OR-specialized open-weight model.

Table 9: Cross-backbone validation of AttackOR. Performance drop is calculated as Original Pass@1 minus Avg. Attacked Pass@1.

Model	Original Pass@1	Avg. Attacked Pass@1	Performance Drop (pp)
Qwen3.5-9B-SFT	65.30	44.79	20.51
Qwen3.5-9B-AttackOR	74.11	72.81	1.30
ORLM-LLaMA-3-8B	59.21	39.85	19.36
ORLM-LLaMA-3-8B-AttackOR	62.75	58.63	4.12

As shown in Table 9, AttackOR improves Avg. Attacked Pass@1 from 44.79% to 72.81% on Qwen3.5-9B and from 40.06% to 58.63% on ORLM-LLaMA-3-8B. The corresponding performance drops decrease from 20.51 to 1.30 percentage points and from 19.19 to 4.12 percentage points, respectively. These results show that the effectiveness of the robustness-training pipeline is not limited to a single model family.

A.3.5 Per-Dataset Robustness Improvement on the Qwen Line

Figure 6 shows the full per-dataset comparison on the Qwen line. AttackOR consistently improves robustness, although the size of the gain differs across datasets and attack families.



Figure 6: Per-dataset robustness improvement on the Qwen line. We compare the SFT baseline and AttackOR on original inputs, each attack family, and the average attacked setting across four datasets.

A.3.6 Dataset-by-Attack Breakdown

Figure 7 visualizes the dataset-by-attack breakdown. The attack impact is clearly dataset-dependent, indicating that robustness patterns cannot be fully characterized by aggregate averages alone.

A.3.7 Evaluation beyond Pass@1

We provide the full comparison between Qwen3.5-9B-SFT and Qwen3.5-9B-AttackOR under the additional metrics introduced in Section 3.5. We also report the mean absolute objective gap, which measures the absolute difference between predicted and ground-truth objective values.

Table 10: Evaluation beyond Pass@1 on the Qwen3.5-9B line. Higher values are better for Pass@k and execution success, while lower values are better for the objective-gap metrics.

Model	Avg. Pass@1	Avg. Pass@3	Avg. Pass@6	Exec. Success	Mean Obj. Gap	Mean Rel. Gap
Qwen3.5-9B-SFT	48.89	62.86	69.81	95.03	65,085,465.38	299.23
Qwen3.5-9B-AttackOR	73.07	74.84	79.15	98.99	13,489,315.79	24.69

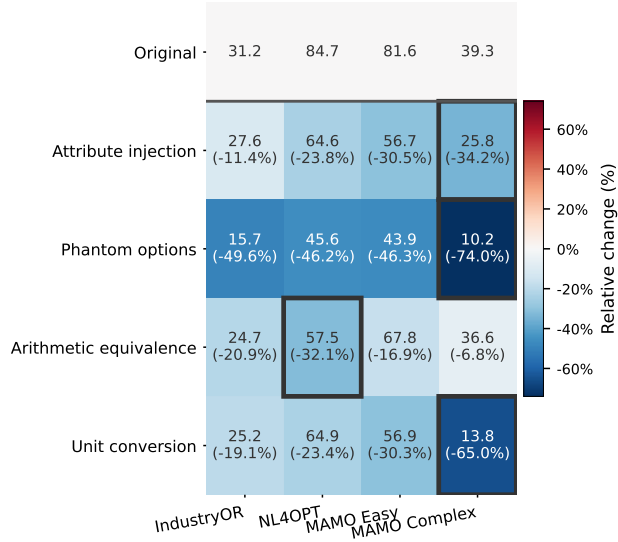


Figure 7: Dataset-by-attack breakdown of attacked performance relative to the original-input setting. Each cell reports Pass@1 under one attack family on one dataset, together with the relative change from the corresponding original-input result.

AttackOR consistently outperforms SFT across Pass@1, Pass@3, Pass@6, and execution success, while reducing the mean relative objective gap from 299.23 to 24.69, as shown in Table 10. These results demonstrate improvements in multi-sample success, code executability, and numerical accuracy.

B LLM Usage

Large language models (LLMs) were used to aid in the writing and polishing of the manuscript. Specifically, we used an LLM to assist in refining the language, improving readability, and ensuring clarity in various sections of the paper. The LLM helped with tasks such as sentence rephrasing, grammar checking, and enhancing the overall flow of the text. It is important to note that the LLM was not involved in the ideation, research methodology, or experimental design. All research concepts, ideas, and analyses were developed and conducted by the authors. The contributions of the LLM were solely focused on improving the linguistic quality of the paper, with no involvement in the scientific content or data analysis. The authors take full responsibility for the content of the manuscript, including any text generated or polished by the LLM. We have ensured that the LLM-generated text adheres to ethical guidelines and does not contribute to plagiarism or scientific misconduct.